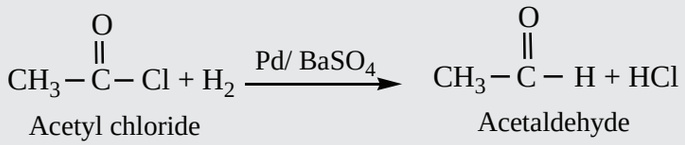
B. Preparation of aldehydes

1) Rosenmund reduction

a) **Aldehydes** can be prepared by the hydrogenation of acid chloride, in the presence of palladium supported by barium sulphate. This reaction is called **Rosenmund reduction**.

Example



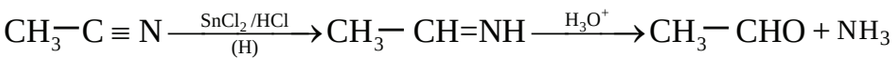
Acetyl chloride — Acetaldehyde

In this reaction, barium sulphate act as a catalytic poison to palladium catalyst, so that aldehyde cannot be further reduced to alcohol.

Formaldehyde and ketones cannot be prepared by this method.

2. Stephen's reaction

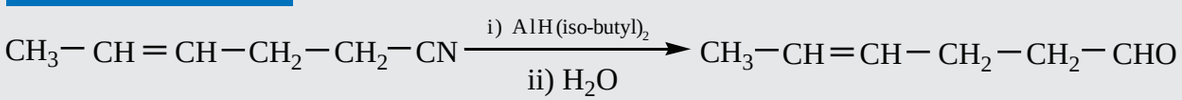
When alkylcyanides are reduced using $SnCl_2/HCl$, imines are formed, which on hydrolysis gives corresponding aldehyde.



3. Selective reduction of cyanides

Diisobutyl aluminium hydride (DIBAL –H) selectively reduces the alkyl cyanides to form imines which on hydrolysis gives aldehydes.

Example

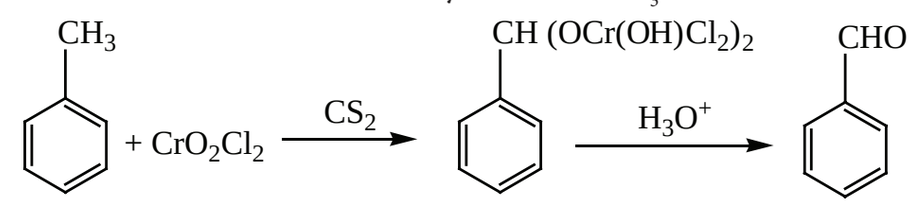


hex - 4- ennitrile — hex - 4- enal

C) Preparation of benzaldehyde

1. Side chain oxidation of toluene and its derivatives by strong oxidising agents such as $KMnO_4$ gives benzoic acid.

When chromylchloride is used as an oxidising agent, toluene gives benzaldehyde. This reaction is called **Etard reaction**. Acetic anhydride and CrO_3 can also be used for this reaction.



methylbenzene — benzaldehyde

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